1st Place: Aspect 12 – Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation

- **Analysis**: This problem cannot be solved in a single step; it requires a series of logical steps to explore and approximate the optimal solution. This process itself is a "multi-step procedure."
- **Manifestation in the Question**: Students likely need to: 1) Understand the constraints and translate them into geometric conditions; 2) Determine the "effective placement area" (the shaded area shrunk inward by 0.5 meters); 3) Analyze the arrangement pattern of circles within this area (e.g., grid or hexagonal packing); 4) Determine the maximum number through calculation or systematic trial; 5) Verify or adjust. This is an exploratory, multi-phase problem-solving process.

2nd Place: Aspect 7 – Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations

- **Analysis**: This is the core of performing concrete operations. Students need to simultaneously handle multiple geometric representations: a gridded floor plan, circles representing the table sets, and a shaded target area.
- **Manifestation in the Question**: Students must measure the dimensions of the shaded area based on the grid, understand the geometric relationship between the circle's diameter (given in the diagram) and the spacing constraints (0.5 meters), and mentally or on sketch paper attempt to arrange the circles. This involves spatial manipulation and numerical processing of geometric shapes (area, circles, distances).

3rd Place: Aspect 13 – Use an understanding of the context to guide or expedite the mathematical solving process

- **Analysis**: This is a highly contextualized problem. Understanding the real-world goal of "customers having enough room" directly translates into the geometric constraint of "0.5 metres away." The context also implies that finding the "maximum number" is an **optimization problem** that may require trying different arrangement strategies.
- **Manifestation in the Question**: Students need to understand that "at least 0.5 metres away" means the **minimum distance** between the **center** of a circle and a wall or the center of another circle is (circle radius + 0.5 meters). This ability to accurately translate textual constraints into geometric conditions relies on deep interpretation of the context. The context also guides the solution goal (maximization), not just simple calculation.

4th Place: Aspect 4 – Make mathematical diagrams, graphs, constructions or computing artifacts and extracting mathematical information from them

- **Analysis**: To find the optimal arrangement, students will likely need to actively construct auxiliary graphical representations.
- **Manifestation in the Question**: Students might **redraw the grid and area to scale on scratch paper**, or **directly attempt to draw circle arrangements on the diagram** (e.g., using grid points as references to plot possible circle center grids). Through this construction and experimentation, they explore possibilities and extract key information: What is the effective "territory" diameter a circle occupies (including itself and its buffer)? How many rows and columns can the effective length and width of the area accommodate?

5th Place: Aspect 6 – Understand and utilise constructs based on definitions, rules and formal systems including employing familiar algorithms

- **Analysis**: Solving this problem requires applying explicit geometric and spatial rules.
- **Manifestation in the Question**: Key rules include:
  - **Properties of a circle** (diameter, radius, center).
  - **Geometric interpretation of distance constraints**: 0.5m from wall to circle edge ⇒ distance from center to wall = circle radius + 0.5m; 0.5m between circle edges ⇒ distance between two centers = circle diameter + 0.5m.
  - **Possible packing knowledge**: Efficiency of square grid vs. hexagonal arrangements. Students need to use these rules to establish a feasible layout model.

6th Place: Aspect 8 – Articulate a solution, showing and/or summarising and presenting intermediate mathematical results

- **Analysis**: Although the question only asks for a final number, arriving at the correct answer requires a clear internal chain of reasoning. At the assessment level, we focus on whether students possess the ability to demonstrate this process.
- **Manifestation in the Question**: A complete solution should be able to explain: 1) The circle's diameter and area dimensions extracted from the diagram; 2) The calculated dimensions of the effective placement area; 3) The chosen arrangement strategy and reason (e.g., "I assumed a square grid arrangement"); 4) The calculation of the number of circles per row and column; 5) The total calculation and possible boundary adjustments. This demonstrates the ability to organize and present intermediate results.

7th Place: Aspect 10 – Make sense of, relate and use a variety of representations when interacting with a problem

- **Analysis**: The problem integrates multiple representations: **text** (constraint descriptions), **2D graphics** (floor plan, circle diagram), **symbols/numbers** (dimensions, distances, final count).
- **Manifestation in the Question**: Students must relate textual constraints ("0.5 metres away") to spatial relationships in the graphics, translate the circle diagram into a specific diameter value, and use the grid scale for actual distance calculations. They need to switch fluently between these representations to construct a solution.

8th Place: Aspect 11 – Switch between different representations in the process of finding solutions

- **Analysis**: This is the dynamic manifestation of Aspect 10, emphasizing the act of switching itself.
- **Manifestation in the Question**: The solving process involves: converting **textual constraints** into **geometric conditions**; converting the **graphical area** into **numerical dimensions**; converting an **arrangement strategy** (a mental model) into a **specific grid point or coordinate layout** (another representation); and finally converting the **layout result** into a **single numerical answer**.

第1位：方面 12 - 使用多步骤流程得出数学解答或结论

- **分析**：解决这个问题无法一步到位，必须通过一系列逻辑步骤来探索和逼近最优解。这个过程本身就是一种“多步骤流程”。
- **题目体现**：学生可能需要：1) 理解约束并将其转化为几何条件；2) 确定“有效放置区域”（阴影区向内收缩0.5米）；3) 分析圆形在该区域内的排列模式（如网格排列或六边形密铺）；4) 通过计算或系统尝试确定最大数量；5) 验证或调整。这是一个探索性、多阶段的问题解决流程。

第2位：方面 7 - 处理数字、图形与几何表示

- **分析**：这是进行具体操作的核心。学生需要同时处理多种几何表示：带网格的平面图、代表桌椅套装的圆形、以及用阴影表示的目标区域。
- **题目体现**：学生必须基于网格测量阴影区域的尺寸，理解圆形直径（由图示给出）与间距约束（0.5米）之间的几何关系，并在头脑中或草图上进行圆形排列的尝试。这涉及对几何图形（区域、圆形、距离）的空间操作和数值化处理。

第3位：方面 13 - 利用对情境的理解来指导或加速求解过程

- **分析**：这是一个高度情境化的问题。对“顾客有足够空间”这一现实目标的理解，直接转化为“0.5米间距”这一几何约束。情境也暗示了寻找“最大数量”是一个**优化问题**，可能需要尝试不同的排列策略。
- **题目体现**：学生需要理解“至少0.5米”意味着圆形**中心**与墙和其他圆形中心之间的**最小距离**是（圆形半径 + 0.5米）。这种将文本约束准确转化为几何条件的能力，依赖于对情境的深度解读。情境还指导了求解的目标（最大化），而非简单计算。

第4位：方面 4 - 制作数学图表、图形或构造物，并从中提取数学信息

- **分析**：为了找出最佳排列，学生很可能需要主动构造辅助的图形表示。
- **题目体现**：学生可能在草稿纸上**重新绘制按比例缩小的网格和区域**，或**在图上直接尝试绘制圆形的排列**（例如，以网格点为参考，画出可能的圆形中心点网格）。他们通过这种构造和试验来探索可能性，并从中提取关键信息：一个圆形占据的有效空间（包括自身和缓冲区）的“势力范围”直径是多少？区域的有效长度和宽度能容纳多少行、多少列？

第5位：方面 6 - 理解并运用基于定义、规则和形式系统的构造

- **分析**：解决此问题需要运用明确的几何和空间规则。
- **题目体现**：关键的规则包括：
  - **圆的性质**（直径、半径、中心）。
  - **距离约束的几何解释**：从墙到圆边缘0.5米 ⇒ 圆心到墙的距离 = 圆半径 + 0.5米；圆形之间边缘距0.5米 ⇒ 两圆心之间距离 = 圆直径 + 0.5米。
  - **可能用到的排列知识**：正方形网格排列或六边形排列的效率。学生需要运用这些规则来建立可行的布局模型。

第6位：方面 8 - 阐明解决方案，展示和/或总结、呈现中间数学结果

- **分析**：虽然题目只要求填写最终数字，但要得出正确答案，学生内部必须有一个清晰的推理链条。在评估层面，我们关注学生是否具备展示此过程的能力。
- **题目体现**：一个完整的解答应能阐述：1) 从图中提取的圆形直径和区域尺寸；2) 计算出的有效放置区域的尺寸；3) 选择的排列策略及理由（如："我假设按正方形网格排列"）；4) 计算每行每列可放置的数量；5) 计算总数并可能进行边界调整。这体现了组织和呈现中间结果的能力。

第7位：方面 10 - 在解决问题时理解、关联和使用多种表征

- **分析**：题目融合了多种表征：**文本**（约束描述）、**二维图形**（平面图、圆形图示）、**符号/数字**（尺寸、距离、最终数量）。
- **题目体现**：学生必须将文本约束("0.5 metres away")与图形中的空间关系关联起来，将圆形图示转化为具体的直径数值，并将网格刻度用于实际距离计算。他们需要在这些表征之间流畅切换以构建解决方案。

第8位：方面 11 - 在求解过程中在不同表征间转换

- **分析**：这是方面10的动态体现，强调转换行为本身。
- **题目体现**：解题过程涉及：将**文字约束**转换为**几何条件**；将**图形区域**转换为**数值尺寸**；将**排列策略**（一种思维模型）转换为**具体的网格点或坐标布局**（另一种表示）；最后将**布局结果**转换为一个**单一的数字答案**。